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Incremental Exercise Test Design and Analysis

Implications for Performance Diagnostics in Endurance Athletes

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Abstract

Physiological variables, such as maximum work rate or maximal oxygen uptake ($\dot{V}O_{2\max}$), together with other submaximal metabolic inflection points (e.g. the lactate threshold [LT], the onset of blood lactate accumulation and the pulmonary ventilation threshold [VT]), are regularly quantified by sports scientists during an incremental exercise test to exhaustion. These variables have been shown to correlate with endurance performance, have been used to prescribe exercise training loads and are useful to monitor adaptation to training. However, an incremental exercise test can be modified in terms of starting and subsequent work rates, increments and duration of each stage. At the same time, the analysis of the blood lactate/ventilatory response to incremental exercise may vary due to the medium of blood analysed and the treatment (or mathematical modelling) of data following the test to model the metabolic inflection points. Modification of the stage duration during an incremental exercise test may influence the submaximal and maximal physiological variables. In particular, the peak power output is reduced in incremental exercise tests that have stages of longer duration. Furthermore, the VT or LT may also occur at higher absolute exercise work rate in incremental tests comprising shorter stages. These effects may influence the relationship of the variables to endurance performance or potentially influence the sensitivity of these results to endurance training. A difference in maximum work rate with modification of incremental exercise test design may change the validity of using these results for predicting performance, and prescribing or monitoring training. Sports scientists and coaches should consider these factors when conducting incremental exercise testing for the purposes of performance diagnostics.

There are a number of biomechanical, environmental, nutritional and psychological factors that will potentially influence endurance performance.^[1,2] However, it is generally regarded that 'performance velocity' (the average speed in an endurance event) will be dictated by the 'performance power' (the average work performed during an en-

durance task) and 'performance oxygen uptake' ($\dot{V}O_2$), which is, in turn, influenced by the percentage of $\dot{V}O_2$ at the lactate threshold (LT), as well as the maximum oxygen uptake ($\dot{V}O_{2\max}$).^[3] These important physiological characteristics are usually determined from incremental exercise testing and

are considered to be necessary for athletes to successfully perform in endurance events.^[3-5]

Incremental exercise testing is a standard procedure for determining submaximal and maximal physiological variables such as $\dot{V}O_{2\max}$ and the LT. However, a key variant in most scientific research and performance diagnosis is the type of incremental test that is used. An incremental protocol can be modified on the basis of the starting work rate, as well as the duration and magnitude of work rate increments. Currently, there is no consensus on the methods used to measure submaximal physiological parameters from such tests. The purpose of this article is to present a contemporary literature review concerning the effects of modifying an incremental exercise protocol on maximal and submaximal physiological variables related to endurance performance. Studies examining the methodological factors influencing the physiological parameters associated with endurance performance will be reviewed. This article focuses specifically on trained endurance athletes.

1. Performance Diagnostics

An aim of sport science research and practice has been to quantify the relationship between maximal and submaximal physiological variables and endurance performance.^[6-11] At the same time, maximal and submaximal physiological variables can be used to prescribe endurance training or to monitor the effects of training.^[12-14] The validity of these physiological variables, in terms of performance diagnostics, are numerous and their interrelationship with each other has not been well defined due to their definitions, incremental exercise protocol design and procedures.

1.1 Definitions and Use of Maximal and Submaximal Physiological Variables

1.1.1 Maximal Aerobic Power

Administering an increase in work rate during incremental exercise has long been used as a method to induce a peak or plateau in $\dot{V}O_2$ (i.e. $\dot{V}O_{2\max}$).^[15,16] Significant correlations between $\dot{V}O_{2\max}$ and distance running, cycling and triathlon performance have been observed in athletes of mixed ability.^[6,7,10,17] However, investigations in elite

athletes have suggested that $\dot{V}O_{2\max}$ does not necessarily distinguish performance in a variety of endurance events.^[4,8,18] At the same time, the relationship between $\dot{V}O_{2\max}$ and endurance performance could be a function of the duration and intensity of the endurance task with shorter duration, higher intensity tasks better correlated with performance.^[10,19] Of more practical significance, the use of $\dot{V}O_{2\max}$ to prescribe training is very limited, as athletes rarely monitor exercise intensity using $\dot{V}O_2$ measurements. Furthermore, it has been commonly observed in well trained endurance athletes that $\dot{V}O_{2\max}$ varies little with short to moderate term training interventions.^[14,20] Hence, while it is important to quantify $\dot{V}O_{2\max}$, it is a variable that may not be that useful in terms of prescribing or monitoring training effects in well trained endurance athletes. Therefore, other maximal or submaximal physiological variables should also be quantified in conjunction with $\dot{V}O_{2\max}$ for the purpose of predicting performance and designing endurance training programmes.

1.1.2 Maximum Work Rate

The maximum work rate obtained during an incremental exercise test has also become popular as a marker of endurance performance in running and cycling.^[10,21-25] Individualised interval training can also be prescribed for cyclists on the basis of peak power output (PPO).^[5,20] In line with this, PPO has also been used to monitor the effects of endurance training in elite cyclists.^[14] If portable devices are available to measure power output during cycling in the field, PPO can be used to determine the intensity of exercise in cycling.^[26] Of practical significance, the PPO has been shown to correlate ($r = 0.97$) with $\dot{V}O_{2\max}$.^[23] Therefore, this variable can be used to predict $\dot{V}O_{2\max}$ without equipment for respiratory exchange analysis. Hence, the maximum work rate obtained from an incremental exercise test to exhaustion is a physiological variable that can be used for a variety of purposes in sport science testing.

There are a variety of terms used to describe the maximum work rate including 'peak power output' or 'maximum work rate' in cycling, or 'peak treadmill velocity' in running.^[22-24] The former definition should not be confused with the PPO obtained during short 'all-out' tests of anaerobic power.^[27] The maximum work rate is obtained by measuring the

highest, fully completed stage for a pre-determined period during an incremental test.^[23] This ranges from 60 seconds to 4 minutes in duration.^[22,28] If a single work stage is not completed, equations can be used to establish the maximum work rate that considers the fraction of the completed stage where fatigue occurred.^[23,29] Therefore, peak work rate is a function of the incremental exercise test design, the implications of which are not well defined.

The reliability (coefficient of variation = 2%) of PPO obtained from an incremental test comprising of short (60-second stages) has been shown to be high.^[30] However, this study is the only one of its kind. Indeed, within the literature, the differences in incremental test protocol are large and this may impact on the maximal (and submaximal) physiological variables measured and the purposes for which they are used. Further studies are required to test the reliability and validity of PPO quantified from incremental tests of different designs.

1.1.3 Submaximal Lactate and Ventilatory Markers

There are a number of submaximal, physiological 'thresholds' or 'inflection' points, which can be quantified using blood lactate or respiratory exchange data collected during an incremental test (see figure 1 and figure 2 for examples). Many of these methods are based on the observation that lactate levels change suddenly at some critical point and, thus, reflect a threshold phenomenon (for reviews,

see Brooks,^[31] Loat and Rhodes,^[32] and Svedahl and MacIntosh^[33]). For example, some authors consider a LT to occur at a work intensity preceding the first increase in lactate concentration above the resting level.^[34] Others have suggested a LT to occur at the preceding of the first increase in lactate concentration of 1 mmol/L.^[35] To overcome the disadvantages of visual, subjective determination of LTs, important changes in lactate concentration may also be identified by using logarithmic transformations^[36] and various curve-fitting procedures, such as the DMax method, which is also thought to correspond with the so-called 'lactate steady state' during incremental exercise.^[37] Other researchers have proposed that a fixed blood lactate concentration (FBLC), such as 2 mmol/L or 4 mmol/L, the so-called 'onset of blood lactate accumulation' (OBLA), can predict endurance performance.^[38] Additional submaximal physiological variables related to the blood lactate response to incremental exercise include the individual anaerobic threshold (IAT), which is determined by exercise and recovery lactate measurements. IAT has been used to quantify an exercise steady state in rowing^[39,40] and reported to induce similar physiological responses during set work exercise to that of the OBLA.^[41] The day-to-day variation (reproducibility) of the corresponding LTs and FBLCs have been shown to be high in trained subjects using an incremental running protocol comprising of 4-minute work stages.^[42] However, more research is required concerning the reliability and agreement of submaximal physiological variables and how specific procedures effect their quantification and relationship.

Submaximal physiological parameters provided by analysis of respiratory exchange data include the pulmonary ventilation threshold (VT),^[43] which can be determined using the expired concentrations of oxygen and carbon dioxide during an incremental exercise task (for reviews, see Davis^[44] and Meyer et al.^[45]). Researchers have also used the terms '1st and 2nd VT' (also termed the 'VT' and 'point of respiratory compensation', respectively) to establish the exercise intensity in endurance events or for the purposes of comparison with non-elite athletes.^[18,46,47] The reproducibility of the 1st and 2nd VTs has been previously presented,^[48-50] with these authors commenting that the 2nd VT is reproduci-

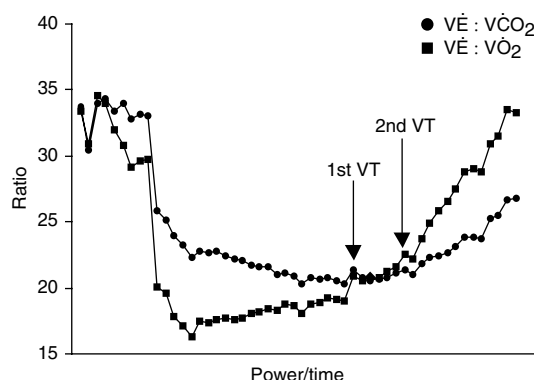


Fig. 1. An example of the 1st and 2nd pulmonary ventilation thresholds (VT) calculated from respiratory equivalents of oxygen uptake ($\dot{V}O_2$) and carbon dioxide release ($\dot{V}CO_2$), together with pulmonary ventilation equivalent ($\dot{V}E$) measured during an incremental exercise test to exhaustion.

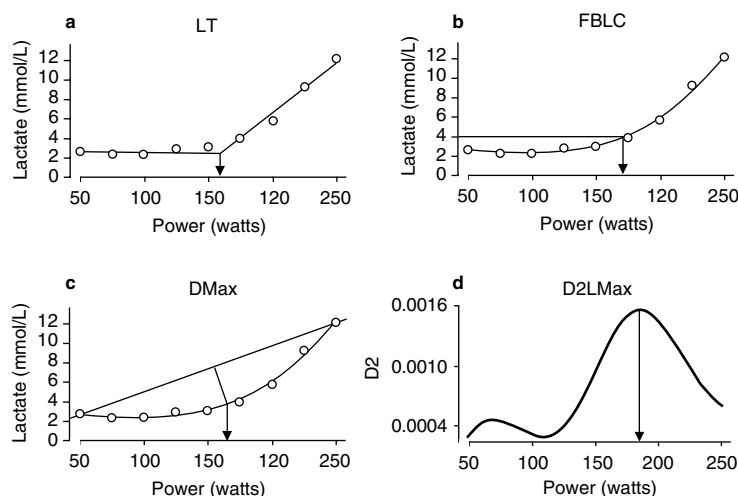


Fig. 2. Examples of submaximal physiological variables obtained from the exercise intensity to blood lactate response obtained from an incremental exercise test. (a) Lactate threshold (LT); (b) fixed blood lactate concentration (FBLC) [= 4 mmol/L]; (c) marker for estimating the LT (DMax LT); and (d) marker representing the point of maximum acceleration in the lactate curve (D2Lmax). D2 = estimated second derivative of the lactate curve.

ble. Given that metabolic acidosis drives pulmonary ventilation, some studies have shown that an inflection in blood lactate concentration above baseline levels is coincident to the 1st VT^[51,52] or the lowest ventilatory equivalent ($\dot{V}E$) for oxygen ($\dot{V}E : \dot{V}O_2$).^[53] However, this has not always been shown^[51,54-56] and is dependent upon the training status of the subjects and the exercise test protocol completed.^[51,57]

It has been demonstrated that submaximal parameters, such as the LT or VT, clearly distinguish endurance performance in athletes with similarly high $\dot{V}O_{2max}$.^[58-60] At the same time, submaximal physiological parameters can be used for prescribing exercise intensity that replicates the intensity experienced during competition.^[61,62] However, it has been shown that the intensity of exercise during a 20km time trial may not agree with that corresponding to the LT.^[63] Others have shown that the blood lactate concentration during a 40km cycling time trial is well above that which corresponds to the LT or FBLCs of 2 and 4 mmol/L.^[64] Indeed, other researchers show that the 2nd VT, which is typically at an exercise intensity higher than the LT, reflects the intensity during a 30-minute cycling time trial.^[65] Another field investigation has shown that in

long duration events, such as the marathon and ironman triathlon, submaximal physiological parameters, such as the 1st VT, correspond well to a race intensity^[66] and equating heart rate (HR) values corresponding to the 2nd VT may over estimate the race intensity during the cycle stage of a half ironman triathlon.^[46] Therefore, whilst the intensity of exercise during events lasting <1 hour seem to correspond well to the 2nd VT, the duration and intensity of an endurance event should be considered before HR or work rate values can be prescribed to simulate a specific race intensity. The coupling of HR values, which correspond to certain submaximal physiological markers, to the race intensity during an endurance event may also be a function of the incremental test used to determine these HR values.

2. The Physiological Effects of Manipulating an Incremental Exercise Test Protocol

2.1 Manipulating an Incremental Exercise Test Protocol

As described in the previous section, incremental tests are often used to determine important

predictors of endurance performance such as the LT, VT, $\dot{V}O_{2\max}$ and PPO or velocity. Therefore, the physiological assessment of the endurance athlete should accurately measure these variables. In studies conducted using trained and untrained populations, a shorter exercise protocol (<60-second stage increments) is typically used to measure $\dot{V}O_{2\max}$ and then, on a second day, a submaximal test is used to quantify the LT and related variables.^[3,67] However, it is also popular to use a single test comprising ≥ 3 -minute stage durations to assess trained subjects.^[8,21,28] Other scientists working with elite cyclists recommend using an incremental test comprising of 60-second stage increments to determine the PPO and VT in cyclists.^[18] It should be highlighted that it is important for an athlete to consume an optimal, high carbohydrate diet for the purpose of inducing full muscle glycogen levels. A number of studies have demonstrated that maximal and submaximal physiological variables or the metabolic response to incremental exercise tasks can be influenced by pre-test glycogen levels or the diet consumed prior to the exercise test.^[68-71] At the same time, acute changes in dietary intake may also influence the metabolic response to incremental exercise and physiological variables calculated from blood lactate values.^[72] However, this response may not be the same for the submaximal physiological parameters determined from respiratory gas exchange measurements.^[70] Hence, the practicing sport scientist has a number of pre-test issues to contend with. However, a sport scientist can approach incremental exercise testing with a variety of protocols aimed at determining a number of different physiological variables. The diversity of incremental exercise testing protocols and the results obtained from these tests has been recently reported.^[73] However, modification of the exercise testing protocol can have implications for the variables measured and, hence, the use of these variables in longitudinal analysis and performance diagnostics.^[74]

An incremental exercise test can be manipulated in a number of different ways to establish the desired physiological adaptation. It is common for the duration (minutes) of each stage of the test, as well as the size of the work increment, to be modified during incremental exercise.^[3,75] Depending upon the exercise mode, the increase may consist of either an

elevation in velocity (m/second) or gradient (percentage) during treadmill tests, or power output (watts) during cycle exercise. Reducing the length of stages or increasing the magnitude of the work required to be performed reduces the total duration of the test as the subject is brought to exhaustion much sooner. The incremental exercise test may also involve a continuous or discontinuous protocol with rest periods between each stage. In terms of blood lactate measurements, the type of blood medium (venous, arterial, mixed arterio-venous) that is obtained may also influence the concentration of this metabolite due to the overall diffusion of lactate into the blood.^[76-81]

2.2 Effects of Manipulating Stage Duration and Increment in Trained Athletes

Incremental exercise test protocols were originally designed to minimise physical discomfort experienced by untrained subjects in order to obtain a valid measure of aerobic capacity and cardiac function.^[15] However, research in untrained subjects suggests that using exercise protocols incorporating stages lasting >3 minutes may compromise the $\dot{V}O_{2\max}$ reached, as well as influence other submaximal physiological parameters.^[81-88] However, the limitation of these studies was that they were conducted with untrained subjects who may respond differently to an incremental exercise task or who are less motivated to continue at higher exercise intensity when compared with trained athletes. Other researchers have reported the effects of manipulating the stage length during an incremental exercise test on maximal and submaximal physiological parameters in trained subjects.^[74,89-92]

2.2.1 Maximum Aerobic Power

There are now a number of studies that have examined whether incremental exercise tests comprising of stages lasting ≥ 3 minutes in duration would result in a lower $\dot{V}O_{2\max}$ in trained cyclists and rowers.^[74,89-92] Pierce et al.^[92] compared four incremental rowing tests using stage durations of either 60 seconds, 3, 4 or 5 minutes. The results showed that there was no difference in $\dot{V}O_{2\max}$ between the 60-second, 3- and 4-minute staged tests. However, the $\dot{V}O_{2\max}$ obtained from the 5-minute stage test was significantly lower. At the same time,

the respiratory exchange ratio was significantly higher in the 60-second stage test when compared with the remaining incremental exercise tests. In another study, McNaughton et al.^[91] demonstrated that $\dot{V}O_{2\max}$ remained unchanged when obtained from an incremental test comprising of 3- or 5-minute stages in cyclists. Similarly, Bishop et al.^[90] found that there was no difference in $\dot{V}O_{2\max}$ between two incremental exercise tests comprising of stage increments of 60 seconds (25 watts) or 3 minutes (10 watts) in duration. Therefore, it appears that, at least in trained athletes, the traditional recommendation of a shorter protocol with large work increments to determine $\dot{V}O_{2\max}$ is not necessarily applicable in this subject group. By performing a test with longer work stages (>3 minutes), rather than shorter stages with rapid increments, valid blood lactate measurements may be obtained. Therefore, other submaximal physiological variables such as the LT or OBLA may be quantified in conjunction with $\dot{V}O_{2\max}$.

2.2.2 Maximum Work Rate

It has been demonstrated that the PPO is lower when the stage duration is increased from 60 seconds to 5 minutes, and that this may reduce the relationship between this variable and endurance performance.^[74,91] In one study, the PPO was not significantly different when measured with incremental tests comprising of 3- and 5-minute stages. In contrast, this study and others^[74,90] have shown that PPO measured during an incremental^[91] exercise test comprising of 60-second stages was significantly higher compared with exercise tests comprising of longer stages.

The implications of employing an incremental test with shorter stage duration or shorter overall length are not well understood in terms of monitoring the effects of training. Furthermore, little is known about the consequences of performing incremental exercise tests of different design and the validity of such protocols for monitoring endurance training. It has been shown in short-term training studies that PPO measured from a longer stage test (2.5-minute stages) is much more sensitive to training induced changes than PPO measured in an incremental tests comprising 60-second stages.^[14,20] Whether this is a function of the type of training or

the incremental test design is not known. However, from these data, it could be hypothesised that an incremental exercise test comprising stages lasting >3 minutes in duration may be more sensitive to performance changes with endurance training. Additional studies are required to compare the results of different incremental tests in athletes exposed to endurance training.

In terms of prescribing training on the basis of peak exercise testing results, it is likely that an incremental test comprising shorter stages will inflate submaximal work rates expressed as a percentage of peak values. Hence, the prescribed work rate may be too severe for an athlete and result in suboptimal acute training responses. In terms of quantifying peak physiological measurements for the purpose of training prescription and monitoring training, it is recommended that an incremental test comprising of longer stages (>3 minutes) may challenge the athlete to a greater extent and may also be more sensitive to changes in field performance. The results may also be more valid in terms of prescribing endurance training intensity; however, this is yet to be confirmed. Future studies are needed to examine: (i) the validity of prescribing endurance training; (ii) the physiological responses to that training; and (iii) tracking of performance improvements using from maximum work rate obtained from different incremental exercise tests.

2.2.3 Submaximal Physiological Parameters

In terms of the blood lactate response to incremental exercise, it has been suggested that it is necessary to use stage lengths of 3–6 minutes during incremental exercise to obtain precise lactate measurements to determine the desired metabolic inflection points.^[35] During submaximal exercise of the same relative intensity, the metabolic response may differ depending upon the training status of the subject being assessed.^[93] In one study, it was shown that the LT was significantly different when obtained from an incremental test comprising of longer (8 minute) stages when compared with a shorter (3 minute) stage test in two groups of subjects with markedly different aerobic capacity.^[94] This result was interpreted as being related to differences in lactate diffusion capacity in the well trained subjects. Yoshida^[79] has also shown that the kinetics

of lactate diffusion may confound blood lactate results during incremental exercise. Thus, the diffusion capacity of lactate and the time allowed (i.e. work duration) for this diffusion to occur before an increment in work rate may influence the blood lactate response to exercise.^[94] This, in turn, could influence submaximal physiological measurements, in particular the exercise intensity corresponding to a 2 or 4 mmol/L FBLC.^[94]

Other research investigations have compared submaximal physiological inflection points obtained from incremental exercise tests comprising different stage durations in trained athletes.^[76,77,95] Weltman et al.^[95] compared the $\dot{V}O_2$, velocity and HR at the LT, and FBLCs of 2, 2.5 and 4 mmol/L obtained from two incremental exercise tests comprising of discontinuous stages of 10 minutes in duration or continuous 3-minute stages. They found that in a group of relatively well trained runners, the speed and $\dot{V}O_2$ corresponding to the LT did not differ between tests. However, the $\dot{V}O_2$ and velocity at a FBLC of 2 mmol/L was significantly higher in the 10-minute discontinuous test. In contrast, Foxdal et al.^[77] concluded that exercise tests using stages of 4–6 minutes in duration do not result in steady-state blood lactate concentrations. Furthermore, these authors suggested that when determining the OBLA threshold in trained subjects, stages of 8 minutes in duration should be used.

Other studies have shown that the work rate corresponding to the VT is different when measured in incremental exercise tests comprising either 60-second or 3-minute stages.^[74] This finding is in contrast with that of Amann et al.,^[89] who showed that there was no difference in the VT when obtained from two tests comprising 60-second or 3-minute stages. Weston et al.^[96] have also shown that an incremental test comprising 'fast' or 'slow' increments resulted in no significant difference in physiological parameters or the VT in trained athletes. However, in both of these studies, the magnitude of the work rate increment was different.

These studies demonstrate inconsistency in terms of manipulating the exercise test protocol and measurement of the VT. More studies are required to examine the validity of measuring the VT using different incremental protocol designs in athletes. These investigations considered, it is recommended

that incremental exercise protocols comprising work stages of >3 minutes in duration may induce more valid blood lactate and respiratory responses. At the same time, incremental exercise protocols comprising stages lasting 3 minutes can be used to measure maximal physiological values and can be coupled to valid submaximal physiological variables. Hence, incremental protocols of this design represent a better approach to holistic performance diagnostics.

Some studies have shown that modifying the length of stages in an incremental test will influence the intensity corresponding to the VT or LT and the relationship of this variable to endurance performance in trained cyclists.^[74,89,97] Despite these studies, there are limited data that have examined the relationship between submaximal physiological variables obtained from incremental tests of different designs and endurance performance. Future research is required to examine whether the incremental exercise protocol influences the relationship between submaximal and maximal physiological parameters, as well as endurance performance.

Modification of an incremental exercise test protocol will influence the LT and OBLA, as well as the peak $\dot{V}O_2$ and the maximum work rate. On the basis of these studies, it also seems that in order to obtain a valid measure of submaximal blood lactate concentrations, longer exercise protocols are needed to allow lactate diffusion before an increment in work occurs. However, using a test comprising longer stages may compromise the $\dot{V}O_{2\max}$ and maximum work rate measurements. While conducting two tests on separate days is one solution, it may be considered too time consuming for athletic populations. In contrast though, more recent work has demonstrated that there may be no difference in the LT and OBLA when obtained from incremental tests comprising stages of 3- or 5-minute duration in well trained cyclists.^[91] Hence, it may be possible to recommend a single incremental testing protocol that is valid for measuring both maximal and submaximal physiological variables.

3. Calculation of Markers of Endurance Performance from Lactate Curves

While the actual incremental protocol used to elicit the lactate response may be crucial in determining the various submaximal inflection points,

the treatment of the blood lactate data generated from an incremental exercise test may also be another important factor. To date, there is no definitive model to describe the blood lactate response to incremental exercise where the emphasis is on modelling the data and not the process that generated the data.

The main controversy surrounding blood lactate analysis is whether there is a breakpoint (i.e. LT) present in the lactate curve or whether lactate increases as a smooth function. It is important to note that the presence of a breakpoint implies a discontinuity in the first derivative of the lactate curve. This assumption does not imply that the lactate curve itself is non-continuous, as highlighted by Morton.^[98]

Traditionally, the LT was determined subjectively from plots of the lactate concentration versus work. Lundberg et al.^[99] proposed fitting a two-part linear regression, where the estimated work rate corresponding to the intersection of the two lines is the LT (i.e. a broken stick regression model). The estimate of LT is found by identifying the LT corresponding to the model with the minimum mean squared error.

Criticisms of the LT marker include that it may be estimating a feature of the curve that does not actually exist and that it is using linear regression, which is quite sensitive to outliers in small datasets (i.e. there can be a considerable difference in the estimate of the LT following small changes in the recorded lactate). Therefore, the use of a line to summarise a curvilinear relationship is questionable. One solution to this problem has been the use of a log transformation of both the work and blood lactate concentration in an attempt to gain a better estimate of the LT.^[36]

The DMax is an alternative objective marker for estimating the LT.^[37] This marker is the intensity corresponding to the point that yields the maximum perpendicular distance from a line joining the first and last lactate measurements to the estimated lactate curve.^[37] The DMax marker represents the exercise intensity where the slope of the line joining the first and last lactate measurements is equal to the slope of the lactate curve (typically estimated using a degree three polynomial). The main criticism of the DMax marker is its dependence on both the

initial and final lactate reading. The initial and final work rate, where the lactate data are collected, may demonstrate considerable variability, which will have a direct influence on the reliability and validity of this marker.^[100] The choice as to where the data collection is stopped is also a vital issue to consider. The marker has been demonstrated to have good reliability and appears to correlate well with the average power output in cycle time trials.^[8,21]

Several other lactate markers, which do not attempt to estimate a breakpoint, have been suggested based on the assumption that during incremental exercise, the change in blood lactate does not display a threshold (i.e. breakpoint) phenomenon. Typically, these markers have no physiological interpretation, but appear to estimate work rates corresponding to points of curvature on the lactate curve.

Endurance markers representing subjective features of a lactate curve have been proposed. One is the intensity corresponding to a FBLC, typically 4 mmol/L,^[101] which represents a ceiling value for lactate. A second, is the intensity preceding an increase in lactate concentration of 1 mmol/L above baseline, where the lactate curve is estimated using a polynomial fit of degree three.^[36] Both of these markers are estimated using inverse prediction. These markers are subjective in nature and, hence, this may influence reproducibility. Smoothing of datasets ensures that the lactate curve is fitted to the data locally, such that regions of considerable variability will not overly influence or 'distort' the estimate of the true curve from that region onwards.

The intensity corresponding to the point of maximum acceleration in the lactate curve has been proposed as a marker.^[102] This marker, the D2LMax, represents a unique feature of the lactate curve and its estimation is not influenced by variability at early and late workloads. The D2LMax is estimated using smoothing splines and is not influenced by initial variation or the choice of end point of the testing protocol. The marker demonstrates good reliability and correlates well with endurance performance.^[100] The D2LMax may be estimated using smoothing splines; however, a simple discrete estimator has also been proposed.^[102] Note, that estimates of the precision of each of the markers (i.e. the standard error) should be considered when determining if changes in a lactate marker are a feature of a real

systematic change in endurance level or an artefact of imprecision in the estimation process.

There does not appear to be a definitive ‘best’ marker in terms of the predictive power of the blood endurance markers and, thus, the LT marker may still be a useful endurance marker while not representing a transition. The correlation between each marker and endurance performance has been assessed using the average power output during a cycle time trial.^[8,10] The correlation of each marker and endurance performance is given in table I, where those markers that were significantly correlated with endurance are flagged. There was no consistent best marker and there was no evidence that those markers that were individually significantly related to performance were indeed significantly different from each other. Multiple linear regression techniques suggested that a selection of markers was more useful than any solitary marker,^[100] allowing for a better prediction of the response of interest (e.g. power output, race time) to be made. However, the use of these markers for prescribing training or determining training-induced changes in performance need to be established.

Interesting research areas in the field of modelling data obtained from incremental exercise testing include: (i) modelling the cumulative effect of lactate on performance (i.e. the area under the lactate

curve); and (ii) using the lactate curve itself (rather than any summary features) for longitudinal comparisons (e.g. across a season) using modern statistical techniques such as Functional Data Analysis (Ramsey and Silverman^[103]).

4. Conclusions

It is typical for researchers to use maximal and submaximal physiological variables to predict endurance performance and design training programmes. A common methodological constraint in all published studies has been the length of the stages during an incremental exercise test and, therefore, the total duration of the incremental test used. At the same time, the approach to analysing a typical blood lactate-to-work rate curve is varied with the validity of the numerous physiological markers not well understood. It is possible that modification of the incremental exercise test protocol and the analysis procedure may affect maximal and submaximal physiological parameters. These changes may be significant in terms of using these physiological variables to monitor the effects of training assessment and for recommendation of exercise or performance prediction. However, there is a limited amount of work in trained populations investigating the reliability and validity of physiological variables obtained from incremental exercise tests comprising of stages of 1–5 minutes in duration. From the available literature, it appears that an incremental test comprising of 3-minute work increments provides the most reliable and valid measures of endurance performance. Future studies in trained athletes are necessary to determine the reliability and validity of maximum work rate, as well as the sensitivity of this variable to training with other submaximal physiological variables measured in different incremental exercise tests.

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Table I. Correlation coefficient of blood lactate endurance markers with power output in cyclists

Lactate endurance marker	Dataset 1 (Bentley et al. 2001) ^[10]		Dataset 2 (Bishop et al. 1998) ^[8]
	Average power output (watts) 20-min TT (n = 9)	Average power output (watts) 90-min TT (n = 9)	Average power output (watts) 60-min TT (n = 24)
LT	0.62	0.76*	0.77*
Log LT	0.66*	0.86*	0.62*
FBLC 4 mmol/L	0.23	0.54	0.80*
1 mmol/L rise above baseline	0.32	0.77*	0.82*
DMax	-0.31	0.35	0.84*
D2LMax	0.64	0.78*	0.76*

DMax = marker for estimating the LT; **D2LMax** = marker representing the point of maximum acceleration in the lactate curve; **FBLC** = fixed blood lactate concentration; **LT** = lactate threshold; **TT** = time trial; * p < 0.05.

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